

NAME:

Practice Problems for Week 9 (due 6/1/26 in class)

Evaluating expressions involving dictionaries

For each of the items given below, say whether they are an expression, a statement, or invalid. Interpret them in the context of variables `d1` and `d2` defined below.

- For expressions, also give their value
- For statements, also say what the effect of the statement is going to be
- For the invalid items, explain why they're invalid

```
d1 = {"tom":344, "anna":643, "peter":9482}
d2 = {"hans":633, "cay":123}
```

(a) `d1["anna"]`

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(b) `d1["hans"]`

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(c) `d1["cay"] = 737`

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(d) `d1 + d2`

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Basic dictionary manipulation

1. Give a statement that creates an empty dictionary and associates it with the variable name `new_dict`

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2. Give an expression that adds a new entry to the (already existing) dictionary `some_dict`. The key of that entry should be 5 and the value should be: `['pokemon', 'call of duty', 'guilty gear']`

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Examining a function

Consider the following function definition.

```
def mystery(alist, blist):  
    if len(alist) != len(blist):  
        return None  
A --> d = {}  
    for idx in range(0, len(alist)):  
B -->     d[alist[idx]] = blist[idx]  
    return d
```

1. What happens in the line marked with A?

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2. What happens in the line marked with B?

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3. In one sentence or short paragraph: what does this function do?

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Dictionaries and variable values

Consider the following program:

```
1. def lists_to_dict(alist, blist):
2.     if len(alist) != len(blist):
3.         return None
4.     d = {}
5.     for idx in range(0, len(alist)):
6.         d[alist[idx]] = blist[idx]
7.     return d
8.
9. new_dict = lists_to_dict(['a', 'b', 'c'], [100, 200, 300])
```

1. Draw a memory diagram that shows what we have in memory once this snippet of code has been completely executed (i.e. after line 9 has been run).
2. Draw a memory diagram that shows what we have in memory when line 4 has just been executed.
3. Draw a memory diagram that shows what we have in memory when line 6 has been executed for the first time.